# Machine Learning Hackathon

**About the Hackathon:**

This hackathon is conducted as part of internal assessment for the Machine Learning (19CSE305) course for B Tech CSE, V Semester students. The aim of the hackathon is to enable & prepare many students of CSE to participate in hackathons being conducted across the Globe in the areas of AI/ML/DS.

The problem statements for the hackathon are carefully chosen and aligned with the grand challenges defined by the University. These 6 grand challenges are detailed in subsequent sections of this document. Each of the problem statement are also aligned with the United Nation's Sustainable Development Goals.

**Venue:** CL-2, CL-3 & CL-4, E Block

**Date:** 18th November, 2024.

**Time:** 8:20 AM – 4:20 PM

**Mode:** Offline; no online participation is allowed.

**Group Size:** 3 - 5 members; students across sections allowed

**Mode of Communication:** MS Teams (WhatsApp group may be formed).

**Important Dates:**

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| Problem Statement Release | 04th November, 2024 |
| Clarification Closure | 06th November, 2024 |
| Group Registration | 09th November, 2024 |
| Solution Proposal Submission | 11th November, 2024 |
| Hackathon Day | 18th November, 2024 |

**Policies:**

- Students from across sections may combine to for groups; each group shall nominate a Team-Lead and Co-Lead for communication purposes.
- Students are allowed to use external tools / sites for references during the hackathon; however, they should refrain from copying content from any 3rd party sources.
- There shall be no specific breaks during the hackathon; students shall plan their own food and coffee breaks.
- Multiple groups are allowed to pick the same problem statement as long as they carry some novelty.
- Groups are not allowed to copy from each other if they have chosen the same problem statement. While such groups are allowed to discuss for ideation and learning, each group shall develop their own solution.

- All materials shall be submitted through TurnItIn or Teams; they shall be subjected to scrutiny for plagiarism.
- All IP created is owned by Amrita Vishwa Vidyapeetham, INDIA. Therefore, no group is allowed to share the solutions or other materials created outside without explicit permission.
- While data sources may be supplied by the organizers, students may opt to use their own data as long as they can justify its usage. All such used data should be legally procured and the License should allow its usage.
- Students are not allowed to withdraw from the event.
- There shall be no alternate evaluation mechanism for students missed the hackathon.
- Additional rules may be brought in as per need.

**Deliverables:**

1. Initial Solution Proposal
2. Final Solution Code
3. Presentation Document / Slides

**Assessment Scheme:** Total Marks = 30 (reduced to 15% contribution)

1. Initial Solution Proposal – 5 Marks (Due: 11th November, 2024) [CO1]
2. Data set & solution understanding – 5 Marks (Due: 18th Nov. 9:30 AM) [CO1]
3. Midway check – 5 Marks (Due: 18th Nov. 11:30 AM) [CO2]
4. Code Review – 5 Marks (Due: 18th Nov. 2:30 PM) [CO2]
5. Final Presentation – 10 Marks (Due: 18th Nov. 4:00 PM) [CO3]

**Assessment Rubrics:**

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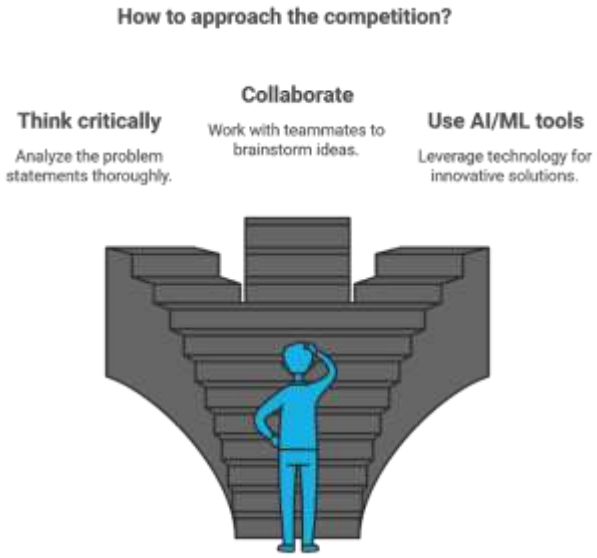
|---|---|
| Dataset Explanation & Initial Understanding | explain how the data is collected, data attributes, why the data is used for the problem statement |
| Midway progress checking for algorithms | proposed solution and algorithms (architecture diagram preferred), justification for selections / choices made |
| code review & validation | Initial code checking, progress so far and course corrections if any needed |
| Hackathon Evaluation: Live Demo / Testing, documentation review | Final system / methodology, Results and comparative analysis, inferences and novelties |

**Grand Challenges of University:**

The six grand challenges defined by Amrita Vishwa Vidyapeetham are:

- One Health (OH)
- Information, Intelligence, & Immersion (II&I)
- Smart, Sustainable, & Resilient Cities & Communities (SSRCC)
- Ecosystem Restoration & Management (ERM)
- Education for Life: Knowledge, Awareness, Interconnectedness (EduLife)
- Empowered Citizens & Organizations (ECO)